Lecture 8: Primary Data Types in C.

Lesson objectives:

By the end of this lesson, you will be able to:

- Define data type
- Describe the information provided by a data type.
- Classify Primary Data Types in C language.
- Choose the right data type to store value.

Primary Data Types in C.

- Data type refer to categorization of values that can be stored.
- In C we can categorize data type into:-
- **a) Primary types:**
  - Integers
  - Floating point
  - Characters
  - Void
- **b) Secondary types**
  - Arrays
  - Enumerated type
  - Structure
  - Typedef
  - Union
  - pointers

Lecture 8: Primary Data Types in C.

Primary/intrinsic types:

i. Integers:

- Used to hold whole numbers.
- Uses keyword **int**
- Uses **%d** format specifier
- Size: Typically 4 bytes (depends on compiler/system).
- Can store values in the range -2^{31} to $2^{31}-1$

ii. Characters

- Used to hold single character.
- Uses keyword **char**
- Uses **%c** as format specifier
- Assigned 1 byte
- Can store values in the range 0 to 255
- Range: signed char: −128 to 127 and unsigned char: 0 to 255

iii. Floating point:

- Used to hold numbers with fractional part.
- Uses keyword **float**
- Size: 4 bytes
- Precision: 6–7 decimal digits
- Uses **%f** as format specifier.
- Can store values in the range −3.4E−38 to 3.4E+38

Lecture 8: Primary Data Types in C.

Primary/intrinsic types:

iv. double:

- Same as float type(decimal numbers)
- Uses keyword **double**
- Assigned 8 bytes
- Precision: 15–16 decimal digits
- Uses **%lf** as format specifier.
- Size: 8 bytes
- Range: −1.7E−308 to 1.7E+308

v. Void

- Type means no value. Keyword used is **void**
- No memory size allocated.
- used to specify the type of functions which returns nothing.

Lecture 8: Primary Data Types in C.

Size modifier- Used to determine increase or decrease of memory size or value range

Modifier	Meaning	Effect on Range	Example
short	Uses less memory	Can store a smaller range of values	Short int x;
long	Uses more memory	Can store a larger range of values	Long int y;

Sign modifier- Used to determine if a variable can store negative values

Modifier	Meaning	Example
Signed	Can store both negative and positive	Signed int y
Unsigned	Can only store positive values	unsigned int x;

Use of data types

Error checking: Data types enable the compiler to **detect mismatches** and improper operations during compilation, helping prevent many logical and runtime errors.

Compatibility: Standardized data types make programs more **portable across** different systems and hardware architectures, ensuring consistent behavior.

Code readability: Clearly chosen data types communicate the nature of the data being handled, making programs easier to understand for both the original developer and others who may read the code.

Code maintainability: Proper use of data types reduces debugging time, minimizes unexpected issues like overflow or precision loss, and makes future updates more manageable.

Lecture 8: Primary Data Types in C

Summary:

- A data type in C is a classification that specifies the type of value a variable can store.
- It defines important information such as the size of memory allocated, the range of values, and the kind of operations that can be performed on the data.
- The primary data types in C include int, char, float, double, and void, which are used to represent basic forms of data.
- Selecting the appropriate data type ensures efficient memory usage and helps prevent errors during program execution.

Lecture 8: Primary Data Types in C.

Activity: You creating a simple app to be used in a local market. After gathering the requirements you have obtained a table data likely to be used in the app. Complete the table below by suggesting the most suitable data type to use, format specifier and a reason for your choice.

No	Data Item	Example Value	Suggested Data Type	Format Specifier	Reason / Justification
1	Quantity of sukuma wiki in kg	50			
2	Price per kg of tomatoes	120.50			
3	Grade of tomatoes	'D'			
4	Discount available (Yes/No)	Yes			
5	Total cost of vegetables	2045.75			
6	Number of customers in a day	300			
7	Average daily rainfall in mm	12.75			
8	Number of market stalls in Nairobi	1500			
9	Number of avocado trees in a farm	120			
10	Weight of a mango in kg	0.25			
11	Price of a bag of maize flour	180			
12	Temperature in Nairobi (°C)	28.5			

Lecture 8: Primary Data Types in C

Exercise:

Describe the following Primary Data Types in C clearly stating the keyword, memory size, possible range of values and format specifiers used

a) Character
b) Integer
c) Floating point
d) Double

Describe the sign modifiers as used in C data types.

Reference:

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